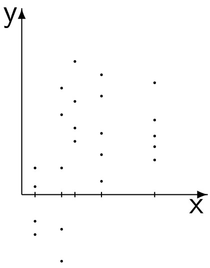
Part 8

Linear Regression
Intercept, Slope
Inference about the Intercept and Slope
F test and ANOVA
Prediction
Lack of Fit of the Linear Model
Correlation
Residual Analysis

LINEAR REGRESSION AND CORRELATION

- Consider *bivariate data* consisting of ordered pairs of numerical values (x, y).
- Often such data arise by setting an x variable at certain fixed values (which we will call levels) and taking a random sample from the population of Y that is assumed to exist at each of the levels of x.
- Here we are thinking of x as not being a random variable, because we are considering only selected fixed values of x (for sampling purposes).
- However, Y is a random variable and we define Y on the population that exists at each level of x.

- Graphically, a scatterplot of the data depicting the y-values obtained by sampling the populations at each of the pre-selected x-values might appear as follows:

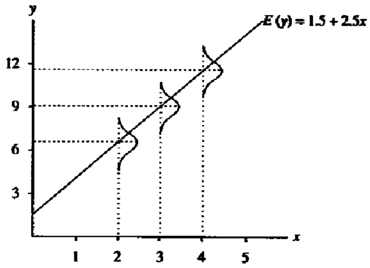


Our objectives given such data are usually twofold:

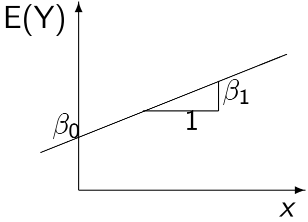
- Summarize the characteristics of the Y populations across values of x – Fit the Model
- Interpolate between levels of X to estimate parameters of Y populations from which samples were not taken – Prediction
- The center of our attention is usually on the means of the Y populations, $E(Y)$, and especially their relationship to one another.
- Considering various relationships among these population means is called parametric modeling.

- *The simple linear regression model* says that the populations at each x-value are normally distributed and that the means of these normal distributions all fall on a straight line, called the *regression line.*

FIGURE 11.2
Theoretical distribution of y in regression



- Let us begin by considering a linear relationship among population means.
- The equation of a straight line through means $E(Y)$ across x-values can be written as $E(Y) = \beta_0 + \beta_1 x$
- Here β_0 is the *intercept* and β_1 is the *slope* of the line.



- The y-values observed at each x-value is assumed to be a random sample from a normal distribution with the mean $E(Y) = \beta_0 + \beta_1 x$, i.e., the mean is a linear function of x.
- The variance of the normal distributions at each x-value is assumed to be the same (or constant).
- Thus the y-values can be related to the x-values through the relationship

$$y = \beta_0 + \beta_1 x + \epsilon \tag{1}$$

- Here ϵ is a random variable (called random error) with mean zero i.e, ($E(\epsilon) = 0$), and variance σ_ϵ^2.
- This model says that sample values are random distances from the line $\mu = \beta_0 + \beta_1 x$ at each x-value.

- The unknown constants in Equation (1), β_0, β_1, and σ_ϵ^2 are called the *parameters* of the model.
- The next question we consider is "How do we proceed to derive a *good* approximating line through Y population means, given only samples from some of the Y populations?"
- In other words, we need to obtain *good* estimates of the parameters of the model using the observed data.
- The phrase *fitting a line through the data* is used to describe our problem.
- It is easy to imagine simply *eye-balling* a line through the points on the scatterplot. But is hard to imagine in what way this can be considered a *good* line.

- The *method of least squares* provides a more sound and clearly defined procedure for fitting a line.
- As an example, consider the data in Section 11.2:

Example: Road Surfacing Data

Project	1	2	3	4	5
Cost y_i(in \$1000's)	6.0	14.0	10.0	14.0	26.0
Mileage (in miles)	1.0	3.0	4.0	5.0	7.0

- In this example, as well as in other examples in this Chapter, for simplicity we will assume that ony one y-value has been observed at each of the x-values.

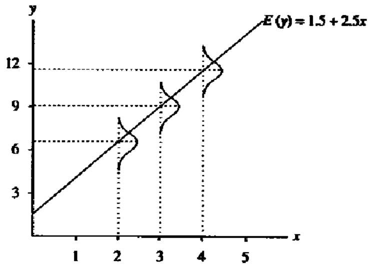
- To explain this method we must first define the terms *predicted value* and *residual*.
- The method of least squares selects a specific line which is claimed to be good. It does so by estimating a value $\hat{\beta}_0$ for β_0 and $\hat{\beta}_1$ for β_1 using the observed data.
- The least squares line then has equation

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where $\hat{y}$ is the point estimate of the mean of the population that exists at x.

- The predicted value $\hat{y}_i$ at a particular value of x_i, is the value of y predicted by the model at that value of x_i i.e., $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$
- The residual is the difference between an observed value y at a given value of x_i, and $\hat{y}_i$ i.e., $(y_i - \hat{y}_i)$.

FIGURE 11.2
Theoretical distribution of y in regression



- The residual $y - \hat{y}$ is the "estimate" $\hat{\epsilon}$ of ϵ, i.e., it is a prediction of the sampling error in y under the assumption that the population means lie on a straight line.
- The method of least squares selects that line which produces the smallest value of the sum of squares of all residuals (hence the name least squares).
- That is, the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to minimize

$$\sum_i (y_i - \hat{y}_i)^2 = \sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

where $(x_i,\ y_i)$ $\ i = 1, 2, \ldots, n$ are pairs of observations.

- Thus $\hat{\beta}_0$ and $\hat{\beta}_1$ are called the least squares estimates (L.S. estimates) of β_0 and β_1, respectively.
- The L.S. estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ given data pairs (x, y) are calculated using the formulae:

$$\hat{\beta}_1 = S_{xy}/S_{xx} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$$

$$\text{where} \quad S_{xx} = \sum(x - \bar{x})^2$$

$$S_{xy} = \sum(x - \bar{x})(y - \bar{y})$$

Handout Example 11.2

After we obtain a least squares fitted line, we are then usually interested in seeing how well the line appears to go through Y population means. One way to investigate this is to look at relative magnitudes of certain sums of squares.

- The Total Variation in all the sample y values is measured by the quantity $\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$
- Let us consider only the numerator

  Total Sum of Squares $= \text{SSTot} = \sum_{i=1}^{n}(y_i - \bar{y})^2$

A little algebraic manipulation will result in the following partitioning of the total sum of squares:

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \sum_{i=1}^{n}\hat{y}_i - \bar{y})^2$$

- We interpret this by noting that the measure of total variation, which is the left part of the equation, is expressible as the sum of two parts which constitute the right side.
- The first part of the right side is the sum of squared residuals Residual Sum of Squares $= \text{SSE} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$.
- We would expect residuals to be close to zero if the Y population means lie close to the estimated least squares line. Thus the smaller the value of SSE closer the regression line will be to the data.
- The other term in the right side of the algebraic identity is Regression Sum of Squares $= \text{SSReg} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$

- Since the left side is the sum of two positive quantities, when SSE decreases SSReg must increase and vice versa.
- The identity is the basis for analysis of variance for regression summarized below:

Source	df	Sum of Squares	Mean Square
Reg	1	SSReg	MSReg=SSReg/1
Error	n-2	SSE	MSE=SSE/(n-2)
Total	n-1	SSTot	

- Later we shall add another column to the above table for calculating an F-statistic for testing a hypothesis about β_1

Interpretation of the Slope Parameter

- In any straight line equation $y = a + bx$, the slope b measures the change in the y-value for a unit change in the x-value (rate of change in y).
- If b is positive y would increase as x increases and if b is negative y would decrease as x increases.
- In the fitted regression model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, the slope $\hat{\beta}_1$ is the change y-value for a unit change in the x-value predicted by the fitted model.

- As in the case when we estimated μ in a single sample case or $\mu_1 - \mu_2$ in the 2 sample case, we need to obtain the standard error of the estimate of $\hat{\beta}_1$ (and of $\hat{\beta}_0$).
- These indicate how accurate our estimates are and help construct confidence intervals and perform tests of hypotheses about the true parameter values β_0 and β_1.

- The standard deviation of $\hat{\beta}_1$, the slope parameter is given by

$$\sigma_{\hat{\beta}_1} = \frac{\sigma_\epsilon}{\sqrt{S_{xx}}}$$

- If the error variance σ_ϵ is large, then $\sigma_{\hat{\beta}_1}$ would be large.
- This says that the slope parameter is estimated with high variability.
- That is, our estimate of the rate of change in y will be less accurate, which will result in, say, a wider confidence interval for $\hat{\beta}_1$.

- By the above formula, we see that the standard deviation of $\hat{\beta}_1$ is also affected by S_{xx}.
- That is smaller S_{xx} the larger $\sigma_{\hat{\beta}_1}$ would be.
- S_{xx} measures the spread of the x-values around its mean. If all the x-values crowd around the mean $\bar{x}$, then S_{xx} will be small.
- In regression this does not help to estimate parameters of the model because the responses to other possible x-values will not be available.
- If we have not selected enough x-values to cover the range of possible y-values we want to predict, then the model we built will not be able to predict changes in those y's with enough accuracy.

- To estimate the above standard deviation we need an estimate of σ_ϵ.
- Since σ_ϵ^2 is the variance of the random errors $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$, we would construct estimate of σ_ϵ^2 based on the residuals $\hat{\epsilon}_i = y_i - \hat{y}_i,\ i = 1, 2, \ldots, n$.
- The estimator of σ_ϵ^2 is

$$s_\epsilon^2 = \frac{\sum_i (y_i - \hat{y}_i)^2}{n-2} = \frac{\mathrm{SSE}}{n-2} = MSE$$

- Recall that for the sample variance s^2 of a sample $y_1, y_2, \ldots, y_n$, we divide $\sum_i (y_i - \bar{y})^2$ by $n-1$ because with $\bar{y}$ we were estimating a single parameter μ.
- In the estimate s_ϵ^2 for the sample variance of the residuals, the divisor is $n-2$ because in $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ we are estimating 2 parameters: β_0 and β_1.
- We say that the Residual SS has $n-2$ degrees of freedom. Using the estimate s_ϵ of σ_ϵ as defined above, the standard error of $\hat{\beta}_1$ is

$$s_{\hat{\beta}_1} = \frac{s_\epsilon}{\sqrt{S_{xx}}}$$

- The standard error of $\hat{\beta}_0$, the intercept parameter is similarly given by

$$s_{\hat{\beta}_0} = s_\epsilon \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

- Thus The standard error of $\hat{\beta}_0$ is also affected by the choice of the x's.
- The intercept estimate is the predicted value of y at $x = 0$.
- In many experimental situations the estimate of the intercept is not of interest as as a value of zero for x is not possible.
- **Refer to the Analysis of Example 11.2: Pharmacy Data**

Computations in the Simple Linear Regression Model

In the textbook, for Example 11.2, the quantities S_{xx}, S_{xy} were computed by first computing the sum of the deviations $(x - \bar{x})$, $(y - \bar{y})$ and the sum of the products $(x - \bar{x})(y - \bar{y})$. In practice, however, the following formulas can be used in hand computations. Thus the computation of the deviations is not necessary:

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n}$$

$$S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n}$$

In Example 11.2, the quantities needed are

- $\sum x = 338, \quad \sum y = 713, \quad \sum x^2 = 14,832\,,$
- $\sum xy = 30,814, \quad \sum y^2 = 64,719, \quad n = 10$
- $S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 14,832 - \frac{338^2}{10} = 3,407.6$
- $S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n} = 30,814 - \frac{(338)(713)}{10} = 6,714.6$
- $S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n} = 64,719 - \frac{713^2}{10} = 13,882.1$

These could be used to obtain the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ as before:

- $\hat{\beta}_1 = S_{xy}/S_{xx} = 6,714.6/3,407.6 = 1.97048$
- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 71.3 - (1.97048)(33.8) = 4.6979$

In addition, the following formulas are needed to compute the quantities for an analysis of variance or ANOVA table:

- $SSTot = S_{yy} = 13,882.1$
- $SSReg = \frac{S_{xy}^2}{S_{xx}} = 13,230.97$
- $SSE = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 13,882.1 - 13,230.97 = 651.13$

This gives the following ANOVA table:

Source	df	Sum of Squares	Mean Square
Regression	1	13,230.97	13,230.97
Error	8	651.13	81.39
Total	9	13,882.1	

Coefficient of Determination:

$$r^2 = \frac{\sum(\hat{y} - \bar{y})^2}{\sum(y - \bar{y})^2} = \frac{\text{SSReg}}{\text{SSTot}} = \frac{13,230.97}{13,882.1} = .9531 \approx 95\%$$

This is a measure of how much better the regression model does in predicting y than just using $\bar{y}$ to predict y.

Inferences about β_0 and β_1

We are still considering the model

$$y = \beta_0 + \beta_1 x + \epsilon,$$

and the least squares fit using a random sample (x_i, y_i), $i = 1, 2, \ldots n$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

is the prediction equation, and the L.S. estimates have form

$$\hat{\beta}_1 = S_{xy}/S_{xx} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \ .$$

- We have assumed that the Y population at each value of x is Normal with mean $\beta_0 + \beta_1 x$
- Each population has the same variance σ_ϵ^2
- Under these assumptions, the least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are each normally distributed:

$$\begin{aligned} \hat{\beta}_1 &\sim N\left(\beta_1, \sigma^2_{\hat{\beta}_1}\right) \\ \hat{\beta}_0 &\sim N\left(\beta_0, \sigma^2_{\hat{\beta}_0}\right) \end{aligned}$$

- We have earlier shown that the estimator of the standard deviation of $\hat{\beta}_1$ is

$$\hat{\sigma}_{\hat{\beta}_1} = s_{\hat{\beta}_1} = \frac{s_\epsilon}{\sqrt{S_{xx}}}$$

- and that the estimator of the standard deviation of $\hat{\beta}_0$ is

$$\hat{\sigma}_{\hat{\beta}_0} = s_{\hat{\beta}_0} = s_\epsilon \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}},$$

- In these formulas, $s_\epsilon = \sqrt{MSE}$

- Using the above results, confidence intervals and tests about the parameters β_1 (and β_0) can be obtained.

A $100(1-\alpha)\%$ Confidence Interval for β_1:

$$\hat{\beta}_1 \pm t_{\alpha/2} \cdot s_{\hat{\beta}_1} \quad \text{giving} \quad \hat{\beta}_1 \pm t_{\alpha/2} \cdot \frac{s_\epsilon}{\sqrt{S_{xx}}}$$

where $t_{\alpha/2}$ is the $100(1-\alpha/2)$ percentile of the student's t distribution with $(n-2)$ degrees of freedom.

Tests of Hypotheses About β_1

Test:

$$
\begin{array}{ll}
H_0: \text{ 1. } \beta_1 \leq 0 & H_a: \text{ 1. } \beta_1 > 0 \\
H_0: \text{ 2. } \beta_1 \geq 0 & H_a: \text{ 2. } \beta_1 < 0 \\
H_0: \text{ 3. } \beta_1 = 0 & H_a: \text{ 3. } \beta_1 \neq 0
\end{array}
$$

Test Statistic: $t = \frac{\hat{\beta}_1 - 0}{s_\epsilon / \sqrt{S_{xx}}}$

Rejection Region: for specified α and $df = n - 1$,

1. Reject H_0 if $t > t_{\alpha,(n-2)}$
2. Reject H_0 if $t < -t_{\alpha,(n-2)}$
3. Reject H_0 if $|t| > t_{\alpha/2,(n-2)}$

where $t_{\alpha,(n-2)}$ is the $100(1-\alpha)$ percentile of the student's t distribution with $(n-2)$ degrees of freedom.

For a hypothesis like $H_0 : \beta_1 = 3$, the test statistic is modified as,

$$t = \frac{\hat{\beta}_1 - 3}{s_\epsilon / \sqrt{S_{xx}}}$$

An F-test from the analysis of variance table

An alternative test of

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_a : \beta_1 \neq 0$$

which is more important in the multiple regression case than our simple linear regression models, comes from the analysis of variance table given below.

Source	df	Sum of Squares	Mean Square	F
Regression	1	SSReg	MSReg	F=MSReg/MSE
Error	n-2	SSE	MSE	
Total	n-1	SSTot		

The F test statistic computed above is used for an F-distribution-based test with $df_1 = 1$ and $df_2 = n - 2$.

Intuitively, large values of this ratio do indicate that the slope β_1 is not zero.

Test: $H_0: \quad \beta_1 = 0$ against $H_a: \beta_1 \neq 0$

Test Statistic: $F = \frac{\text{MSReg}}{\text{MSE}}$

Rejection Region: Reject H_0 if $F > F_\alpha$

where F_α is the $100(1-\alpha)$ percentile of the F distribution with $df_1 = 1$ and $df_2 = n - 2$

Example 11.6

A simple linear regression model was fitted to the mean age, x, of executives of 15 firms in the food industry and the previous year's percentage increase in earning per share of the firms, y.

Mean Age	38.2	40.0	42.5	43.4	44.6	44.9	45.0	45.4
% Change(in earnings per share)	8.9	13.0	4.7	-2.4	12.5	18.4	6.6	13.5
Mean Age	46.0	47.3	47.3	48.0	49.1	50.5	51.6	
% Change(in earnings per share)	8.5	15.3	18.9	6.0	10.4	15.9	17.1	

- The quantities needed for the computation are
  $\sum x = 683.8, \quad \sum y = 167.3, \quad \sum x^2 = 31,358.58$
  $\sum xy = 7,741.74, \quad \sum y^2 = 2,349.61$

- Using these it follows that
$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 31,358.58 - \frac{683.8^2}{15} = 186.4173$
$S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n} = 7,741.74 - \frac{(683.8)(167.3)}{15}$
$= 115.0907$
$S_{yy} = \sum y^2 - \frac{(\sum y)^2}{n} = 2,349.61 - \frac{167.3^2}{15} = 483.6573$
- These could be used to obtain the estimates $\hat{\beta}_0$, $\hat{\beta}_1$, and $s_\epsilon = \sqrt{MSE}$ as before.

- The calculations are:
  $\hat{\beta}_1 = S_{xy}/S_{xx} = 115.0907/186.4173 = 0.617382$
  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 11.153 - (0.617382)(45.5867) = -16.991$
  $SSE = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 483.6573 - \frac{115.0907^2}{186.4173} = 412.60236$
  $MSE = SSE/(n-2) = 412.60236/13 = 31.7386$
  $s_\epsilon = \sqrt{MSE} = 5.634$
- A 95% confidence interval for β_1 is: $\hat{\beta}_1 \pm t_{.025,13} \cdot \frac{s_\epsilon}{\sqrt{S_{xx}}}$

- It is calculated as:

$$0.617382 \pm (2.16)(\frac{5.634}{\sqrt{186.4173}}) \quad \text{or} \quad 0.617382 \pm 0.89130$$

  i.e., $(-0.27392, 1.5087)$.

- In this problem, to determine if executive age has any predictive value for predicting change in earnings, we need to test $H_0: \ \beta_1 = 0 \quad \text{vs.} \quad H_a: \ \beta_1 \neq 0$

- We chose the two-sided research hypothesis because, if executive age was a good predictor, we do not know whether it would have a negative or a positive effect on change in earnings. We use $\alpha = .05$ for the test.

Test Statistic:

$$t_c = \frac{\hat{\beta}_1 - 0}{s_\epsilon/\sqrt{S_{xx}}} = \frac{(0.617382 - 0)}{5.634/\sqrt{186.4173}} = \frac{0.617382}{0.412642} = 1.496$$

Rejection Region:

$$|t| > t_{.025,13} = 2.16$$

- Since the computed t-statistic t_c is not in the rejection region we fail to reject H_0. Thus there is no evidence to conclude that change in earnings can be modeled using executive age as a predictor in a simple linear regression model.

- We can also use an F-test to test the above hypothesis. The calculations above gives the following Anova table:

Source	df	Sum of Squares	Mean Square	
Regression	1	71.0549	71.0549	2
Error	13	412.6024	31.7386	
Total	14	483.6573		

- The rejection region for the F-test at $\alpha = .05$ is $F > F_{.05,1,13}$ i.e., $F > 4.67$ from Table 8.
- We fail to reject $H_0: \ \beta_1 = 0$ at $\alpha = .05$ as F_c is not in the R.R.
- From the F table, the p-value is between .10 and .25.

- Coefficient of determination is

$$r^2 = \frac{\text{SSReg}}{\text{SSTot}} = \frac{71.0549}{483.6573} = .1469 = 14.7\%$$

- This says that using executive age as a predictor of change in earnings in a straight line model is only 14.7% better than using the sample mean of change in earnings.
- Another interpretation of r^2 is that it is the proportion or percentage of variation in y that is explained by $\hat{y}$. In multiple regression models, this interpretation is affected the number of x variables in the model.

Predicting New y Values Using Regression

- There are two possible interpretations of a y prediction at a specified value of x.
- Recall that the prediction equation for the highway construction problem was $\hat{y} = 2.0 + 3.0\,x$, where $y =$ cost of highway construction contract and $x =$ miles of highway.

  at a specified value of x.
- The highway director substitutes $x = 6$ in this equation and gets the value $\hat{y} = 20$.
- This predicted value of y can be interpreted in one of two ways.

- The predicted value $\hat{y} = 20$ can be interpreted as either

  The average or mean cost $E(y)$ of all resurfacing contracts for 6 miles of road will be \$20,000.

  or

  The cost y of a specific resurfacing contract for 6 miles of road will be \$20,000.

- The difference in the two predictions is that the standard error of predictions are different. Therfore, the confidence intervals associated with each of them will also be different.
- Since it is easier to more accurately predict a mean than an individual value, the first type of prediction will have less error than the the second type.

Predicting the mean $E(Y)$ at a given x

For any Y population, $E(Y)$ is the population mean. According to our model, the expression for $E(Y)$ in terms of x and the parameters β_0 and β_1 is

$$E(Y) = \beta_0 + \beta_1 x \,.$$

Note that this is linear function of the parameters β_0 and β_1.

The least squares estimate (i.e., the point estimate) of $E(Y)$ for a given population at a new value of x (call it x_{n+1}) is

$$\hat{y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 x_{n+1} \,.$$

Using our assumptions about ϵ in the model description, the standard deviation of $\hat{y}_{n+1}$ is

$$\sigma_\epsilon \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

The estimate of this, called the standard error of $\hat{y}_{n+1}$ is:

$$\text{s.e.}(\hat{y}_{n+1}) = s_\epsilon \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

where $s_\epsilon^2 = \mathrm{SSE}/(\mathrm{n} - 2)$

Since we assume normally distributed data we have that a $100(1-\alpha)\%$ confidence interval for $E(Y)$ is

$$\hat{y}_{n+1} \pm t_{\alpha/2} \cdot s_{\epsilon} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

where $t_{\alpha/2}$ is based on $df = n-2$.

Example: (Example 11.2 continued)

- The prediction equation in the pharmacy example is $\hat{y} = 4.70 + 1.97\,x$.
- If the % of ingredients purchased directly by a pharmacy is 15, i.e., $x_{n+1} = 15$, obtain a 95% confidence interval for the mean sales volume $E(Y_{n+1})$ for similar pharmacies.

- The point estimate of $E(Y_{n+1})$ at $x_{n+1} = 15$ is $\hat{y}_{n+1} = 4.70 + (1.97)(15) = 34.25$ as we have seen before.
- The 95% confidence interval for the mean sales volume at $x_{n+1} = 15$ is

$$\hat{y}_{n+1} \pm t_{.025,8} \cdot s_{\epsilon} \sqrt{\frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

$$34.25 \pm (2.306)(9.022)\sqrt{\frac{1}{10} + \frac{(15 - 33.8)^2}{3407.6}}$$

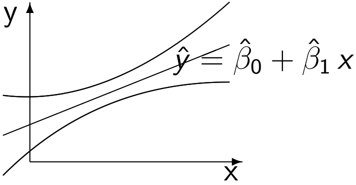
$$34.25 \pm 9.39$$

- This gives (24.86, 43.64) or ($\$24,860$, $\$43,640$) as the 95% confidence interval for $E(Y_{n+1})$ at $x_{n+1} = 15$.
- The confidence interval for $E(Y)$ becomes wider as x_{n+1} gets further away from $\bar{x}$ because the term

$$\frac{(x_{n+1} - \bar{x})^2}{S_{xx}}$$

gets larger. This is called the extrapolation penalty.

- Since the above interval has endpoints that are a function of x_{n+1} it yields a $100(1-\alpha)\%$ confidence band for $E(Y)$ at all possible x_{n+1} values.



- Note that the interval is narrowest at the point $x = \bar{x}$ and gets wider as x move away from $\bar{x}$ and the prediction becomes less accurate.

Predicting a future observation y at a given x

- Often it is more relevant to ask a question like "If I take an observation at $x = x_{n+1}$, what y value am I likely to get?"

  In other words we are asking what y should we predict at $x = x_{n+1}$.

- This is different from estimating the average (mean) $E(Y)$ at $x = x_{n+1}$.

- We now want to predict the value of a future observation, not estimating the population mean $E(Y)$ at $x = x_{n+1}$.

- The least squares prediction of y at a new value of x_{n+1} is

$$\hat{y}_{n+1} = \hat{\beta}_0 + \hat{\beta}_1 \, x_{n+1} \, .$$

- This is the same as the estimate of $E(Y)$.
- However, the standard error of prediction is different.
- We are estimating $\beta_0 + \beta_1 x$ but predicting y, i.e., $y = \beta_0 + \beta_1 x + \epsilon$, so $\mathrm{Var}(\epsilon) = \sigma^2_\epsilon$ must be accounted for.
- As we did for a confidence interval for $E(Y)$ we can derive a prediction interval for the future y_{n+1}.

A $100(1-\alpha)\%$ Prediction Interval for a future y_{n+1} at x_{n+1} is

$$\hat{y}_{n+1} \pm t_{\alpha/2} \cdot s_\epsilon \sqrt{1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

where $s^2_\epsilon = \mathrm{SSE}/(\text{n-2})$, and $t_{\alpha/2}$ is based on $df = n - 2$.

- Note that a 1 has been added to the square root part of the standard error of $\hat{y}_{n+1}$.
- This represents the addition of an extra s_ϵ to the standard error.
- This means that there is greater error in predicting a future observation y_{n+1} compared to estimating a mean $E(y)$, as discussed earlier.

Example: (Example 11.2 continued)

If the % of ingredients purchased directly by a pharmacy is 15, i.e., $x_{n+1} = 15$, obtain a 95% prediction interval for the sales volume y for that pharmacy.

- The 95% prediction interval for the sales volume at $x_{n+1} = 15$ is

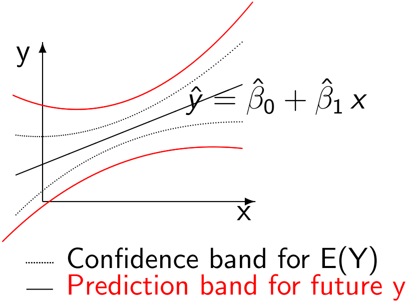
$$\hat{y}_{n+1} \pm t_{.025,8} \cdot s_\epsilon \sqrt{1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}}}$$

$$34.25 \pm (2.306)(9.022)\sqrt{1 + \frac{1}{10} + \frac{(15 - 33.8)^2}{3407.6}}$$

$$34.25 \pm 22.83$$

- That is, $(11.43, 57.08)$ or $(\$11,430, \$57,080)$.
- As you will notice this is a much wider interval than the 95% confidence interval for $E(Y)$ the mean sales volume at $x_{n+1} = 15$.

- Since the endpoints of the above prediction interval are a function of x_{n+1}, this is actually a prediction band. This band will contain the confidence band for y_{n+1}.



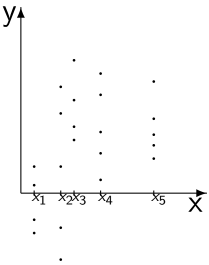
A Statistical Test for Lack of Fit of the Linear Model

- The assumptions we have made about the distribution of ϵ's in our linear regression model permit us to derive a test for lack of fit under certain conditions which we will describe.
- Whenever the data contain more than one observation at one or more levels of x, we can partition SSE into two parts.
- This is another algebraic identity like we have seen for partitioning total variability into SSReg and SSE.
- Let the data be now represented as:

$$(x_i,\ y_{ij}) \quad i = 1, 2, \ldots, k; \quad j = 1, 2, \ldots, n_i$$

- n_i is the number of observations taken at x_i

- Thus we imagine k levels of x and at each x_i there are n_i observations $y_{ij}, j = 1, 2, \ldots, n_i$.
- Graphically we envision a situation like the one shown below.



- Note that n_i may be equal to 1 in some cases. If $n_i = 1$ for all x_i then we have no repeated observations at any of the x's and we cannot test for lack of fit.
- The algebraic identity is:

$$\sum_i \sum_j (y_{ij} - \hat{y}_i)^2 = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2 + \sum_i \sum_j (\bar{y}_i - \hat{y}_i)^2$$

SSE (Sum of squares of residuals)	$\mathrm{SSE}_{\mathrm{exp}}$ (SS due to pure experimental error)	$\mathrm{SS}_{\mathrm{Lack}}$ (SS due to lack of fit)

- Note that the last term in the right hand side of the above equation is $\sum_i \sum_j (\bar{y}_i - \hat{y}_i)^2$
- If indeed there is a linear relationship $E(Y) = \beta_0 + \beta_1 x$ among Y population means, then this sum of squares should not be large.
- That is because $\bar{y}_i$ is a point estimate of $E(Y_i)$ at x_i, and $\hat{y}_i$ is a point estimate of the same mean $E(Y_i) = \beta_0 + \beta_1\, x_i$.
- The hypotheses we might test are:

$$H_0 : \text{A linear model is appropriate vs.}$$

$$H_a : \text{A linear model is not appropriate}$$

- The test for lack of fit is an F test.

- The F statistic is the ratio of mean squares for lack of fit and mean squares for pure experimental error $\sum\sum(\bar{y}_i - \hat{y}_i)^2$
- The mean squares are sums of squares divided by their degrees of freedom (this is the definition of a mean square).

$$\mathrm{MS}_{\mathrm{Lack}} = \frac{\mathrm{SS}_{\mathrm{Lack}}}{(n-2) - \sum_i (n_i - 1)} \equiv \frac{\sum_i \sum_j (\bar{y}_i - \hat{y}_i)^2}{(n-2) - \sum_i (n_i - 1)}$$

$$\mathrm{MS}_{\mathrm{exp}} = \frac{\mathrm{SSE}_{\mathrm{exp}}}{\sum_i (n_i - 1)} \equiv \frac{\sum_i \sum_j (y_{ij} - \bar{y}_i)^2}{\sum_i (n_i - 1)}$$

- The F statistic is: $F = \frac{\mathrm{MS}_{\mathrm{Lack}}}{\mathrm{MS}_{\mathrm{exp}}}$
- We reject H_0 at level α whenever the computed value of the F-statistic exceeds F_{α,df_1,df_2} (i.e. the computed F-statistic is in the rejection region).
- F_{α,df_1,df_2} is the $100(1-\alpha)$ percentile from the F table with degrees of freedom $df_1 = n - 2 - \sum_i (n_i - 1)$ and $df_2 = \sum_i (n_i - 1)$.
- Failure to reject H_0 implies that there is not enough evidence to declare that the linear model is not appropriate.

Correlation

- We have proceeded under the assumption that Y population means fall on a straight line.
- We computed the least squares line as an approximation to this straight line.
- We also looked at the sum of squared residuals as an indicator of relative success in explaining variation in Y.
- There is a measure of the strength of the linear relationship between two variables X and Y. It is called correlation coefficient ρ.
- ρ is a parameter associated with the bivariate distribution of X, Y (much like μ or σ^2 for a univariate distribution).

- The estimate of ρ is called the sample correlation coefficient r.
- For n pairs of observations (x_i, y_i) we define

$$
\begin{aligned}
r &= \frac{S_{xy}}{\sqrt{S_{xx}\,S_{yy}}} \\
\text{where} S_{xx} &= \sum (x_i - \bar{x})^2, \quad S_{yy} = \sum (y_i - \bar{y})^2 \\
S_{xy} &= \sum (x_i - \bar{x})(y_i - \bar{y})
\end{aligned}
$$

Properties of r are:

- $-1 \leq r \leq 1$.
- $r = 0$ indicates no linear relationship between x and y.
- $r = 1$ indicates a perfect linear relationship between x and y, and the line has positive slope.
- $r = -1$ also indicates a perfect linear relationship, but with negative slope.
- Strength is measured, relatively, by how far $|r|$ is from 0 and 1.
- We imagine a true correlation existing as the unknown value of the parameter ρ, and r is the estimate of ρ based on a sample.

- It can be shown that

$$r^2 = \frac{\sum(\hat{y} - \bar{y})^2}{\sum(y - \bar{y})^2}$$

- i.e. r^2 is the ratio of the sum of squares due to regression to the total sum of squares. This is the same as the coefficient of determination defined earlier.
- Its interpretation is that r^2 is the proportion of total variability in Y accounted for by the model.
- Since r only indicates the strength of the linear relationship between x and y, its value is not useful when there is a strong curved relationship.

Diagnosing the Fitted Model: Residual Analysis

- We review first the consequences of the assumptions of normality, homogeneity of variance, and independence of errors ϵ_i in the model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $i = 1, 2, \ldots, n$.
- Recall that each y is a normal random variable because it equals a constant plus a normal random variable, and that the y_i's are independent because the ϵ_i's are independent.
- We also assumed that the variances of the ϵ_i's for the populations at each of the x_i's are the same and is equal to σ_ϵ^2. This is called the homogeneity of variance assumption.

The consequences of this are the results concerning the distributions of $\hat{\beta}_1$, $\hat{\beta}_0$, and $\hat{y}_i$ that we have already used in the inference procedures so far discussed.

That is

$$t = \frac{\hat{\beta}_0 - \beta_0}{s_\epsilon \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}},$$

$$t = \frac{\hat{\beta}_1 - \beta_1}{s_\epsilon \sqrt{S_{XX}}},$$

and,

$$t = \frac{\hat{y} - E(y)}{s_\epsilon \sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{XX}}}}$$

are each distributed as a Student's t random variable.

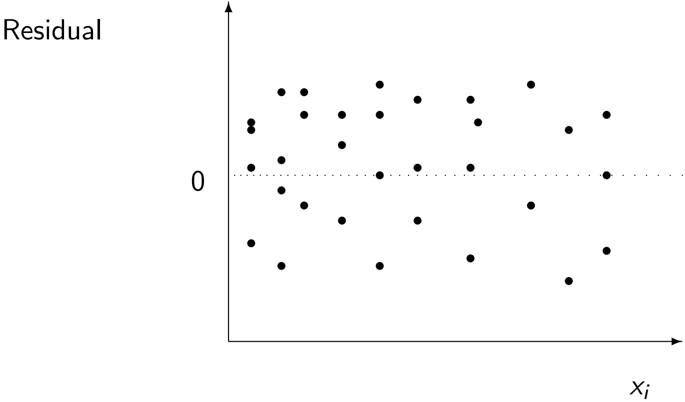
Residual plotting to look for possible violation of assumptions about ϵ

Graphics can be used to examine the validity of the assumptions made about the distribution of ϵ's. These plots are based on the fact that the residuals from fitting the model, $e_i = y_i - \hat{y}_i$ for $i = 1, \ldots, n$ reflect the behavior of the ϵ's in the model.

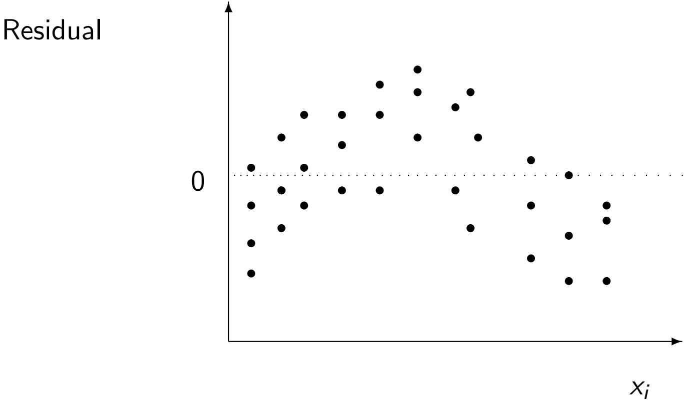
Plot of $\boldsymbol{e_i}$ vs. $\boldsymbol{x_i}$.

- If the model is correct, we would expect the residuals to scatter evenly and randomly around zero as the value of x_i changes.
- If a curved or nonlinear pattern is shown, it indicates a need for higher order or a nonlinear terms in the model.

- This plot may also show a pattern if there are outliers present.
- This plot may also show violation of the homogeneity of variance assumption if the variance depends on the actual value of x_i. It will show up as a marked decrease or increase of the spread of the residuals around zero.



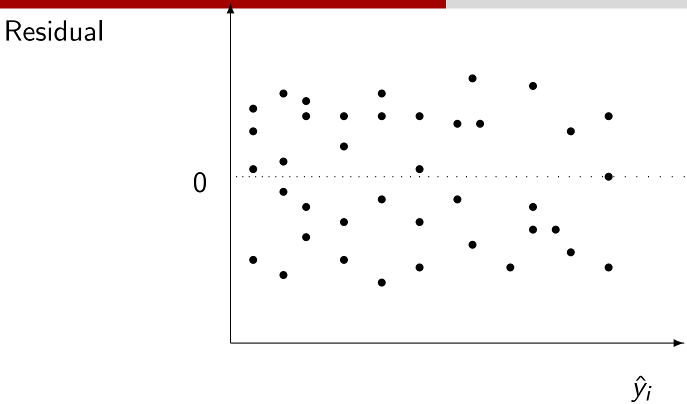
No Pattern



Pattern

Plot of e_i vs. the predicted values $\hat{y}_i$

- This scatterplot should show no pattern, and should indicate random scatter of residuals around the zero value.
- A pattern indicating an increase/decrease in spread of the residuals as $\hat{y}_i$ increases shows a dependence of the variance on the mean of the response. Thus the homogeneity of variance assumption is not supported.



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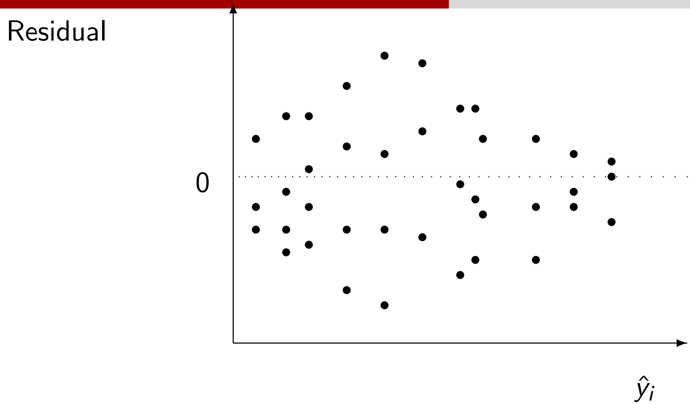
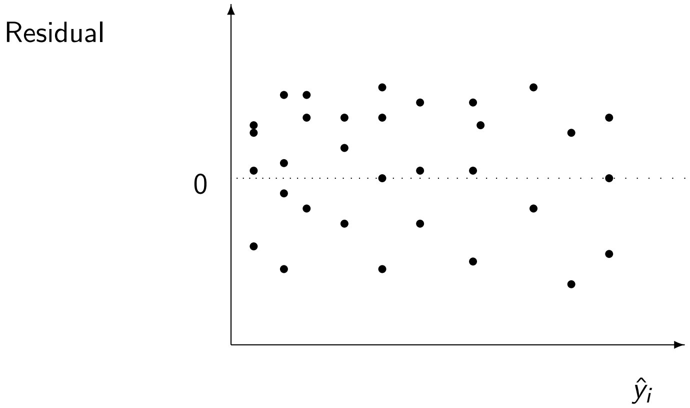
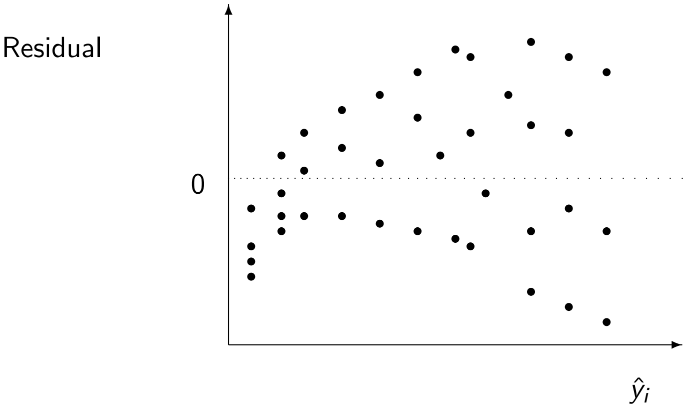


Pattern

The above kind of spread pattern may also show up along with the curvature pattern in both this and the previous plot if higher order terms are needed, as well.



No Pattern



Pattern

Normal probability plot of the studentized residuals

- This plots quantiles of a standardized version of residuals against percentiles from the standard normal distribution.
- The points will fall in an approximate straight line if the normality assumption about the errors (ϵ's) is plausible.
- Any other pattern (as discussed earlier) will indicate how this distribution may deviate from a normal distribution.
- For example, a cup-shape indicates a right-skewed distribution while a heavy-tailed distribution is indicated by a reverse S shape.
- This plot may also identify one or two outliers if they stand out from a well-defined straight line pattern.